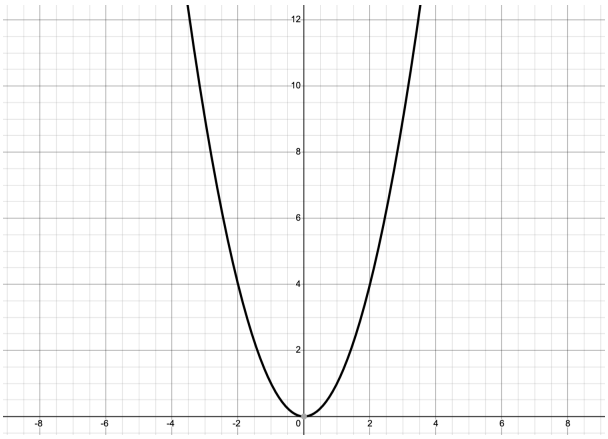
So far, all of our optimization problems could be reduced to finding the absolute max or min of a continuous function on a closed interval. However, many interesting optimization problems require us to find an absolute max or min of a continuous function that is defined on an interval which is not closed. For such functions, the Extreme Value Theorem does not guarantee the existence of an absolute extremum, but, as we have seen, one might exist nonetheless.

Example. The function $f(x) = x^2$ on $(-\infty, \infty)$ has no absolute maximum, but does have an absolute minimum of 0 at $x = 0$.



Since the Extreme Value Theorem does not apply to such functions, we cannot just make a table of critical points and check their values as we did before, as it is possible that the function has local extrema but no absolute extrema. However, if the function has a single critical point, which is often the case in constrained optimization, then we have the following result:

Proposition. Suppose f is a continuous function defined on an interval I, which need not be closed. If f has a local extremum at x_0 and no other critical points, then $f(x_0)$ is a global extremum.

Exercise. Prove this proposition in the case of a local maximum (the proof for the case of a local minimum is similar). (*Hint:* Assume $f(x_0)$ is a local maximum but not a global maximum. Then there exists x_1 such that $f(x_1) > f(x_0)$. Use the Extreme Value Theorem on the interval with endpoints x_0 and x_1 to derive a contradiction.)

Examples

1. A rectangular box with a square base, an open top, and a volume of 216 in^3 is to be constructed. What should the dimensions of the box be to minimize the surface area of the box? What is the minimum surface area?

2. We want to construct a box whose base length is 3 times the base width. The material used to build the top and bottom cost \$10 / ft^2 and the material used to build the sides cost \$6 / ft^2. If the box must have a volume of 50 ft^3 determine the dimensions that will minimize the cost to build the box.

3. A manufacturer needs to make a cylindrical can that will hold 1500 cm^3 of liquid. Determine the dimensions of the can that will minimize the amount of material used in its construction. (Recall that $V_{\text{cyl}} = \pi r^2 h$ and $A_{\text{cyl}} = 2\pi rh + 2\pi r^2$.)

4. A printer needs to make a poster that will have a total area of 200 in^2 and will have 1 in margins on the sides, a 2 in margin on the top, and a 1.5 in margin on the bottom. What dimensions will give the largest printed area?